EUonAIR / RESILIENT FUTURES LAB

What can 120 hryvnias buy?



Lunch prices, purchasing power and inflation

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AT THE SCHOOL GATE

Marta has 120 UAH for lunch

She visits the same bakery and buys the same kind of roll on both days.

ON MONDAY



30 UAH

per roll

ON FRIDAY



40 UAH

per roll

ON YOUR OWN
1 MIN

How many rolls can Marta buy on each day? What changed for her?

UAH is short for Ukrainian hryvnia.

One fewer roll for the same money

WHAT 120 UAH BUYS

At 30 UAH



At 40 UAH



Marta still has 120 UAH. Her lunch budget buys three rolls instead of four.

WHY PEOPLE ACCEPT MONEY

The baker can spend it elsewhere. Marta can use it to compare prices or save it for later.

Purchasing power is how much that money can buy.

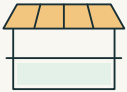
Who decides what a roll costs?



Demand

How many rolls would students buy at each price?

They weigh the price against their budget. Some bring lunch from home when rolls cost more.



Supply

How many rolls would the stalls sell at each price?

Each extra batch takes flour, oven time and labour. A higher price can make that batch worth baking.

Classroom market with competing stalls. Federal Reserve Bank of St. Louis, how prices form.

A LUNCHTIME EXPERIMENT

Which price would clear the stalls?

Here is what students would buy and the stalls would bake for one lunchtime.

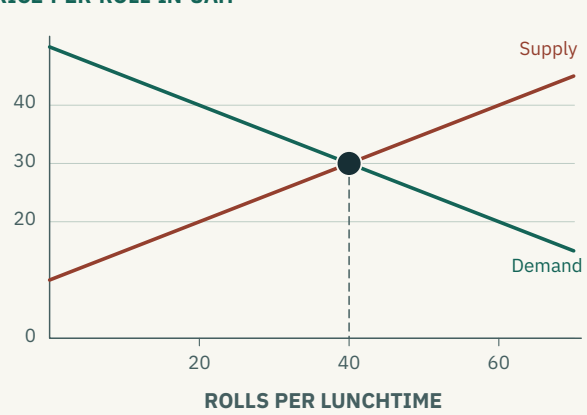
PRICE PER ROLL	ROLLS STUDENTS WANT	ROLLS STALLS OFFER
20 UAH	60	20
30 UAH	40	40
40 UAH	20	60

IN PAIRS
2 MIN

Which price matches buyers and sellers? How many buyers miss out at 20 UAH?

Assume each customer wants one roll and the stalls can adjust prices.

At 30 UAH, 40 rolls change hands



AT 20 UAH

A shortage of 40 rolls leaves a queue. A stall could raise its price.

AT 40 UAH

A surplus of 40 rolls gives sellers a reason to cut the price.

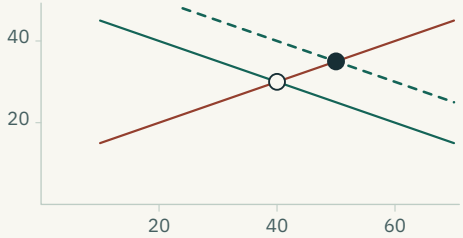
At equilibrium, the two quantities match.

Lines connect the quantities and extend the same pattern. St. Louis Fed, market prices.

Why might a roll rise to 35 UAH?

A school match draws a crowd

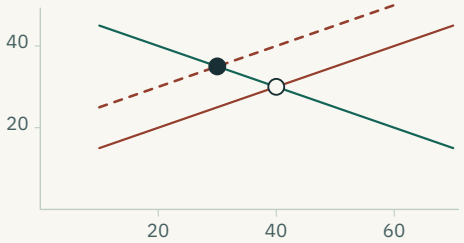
At each price, students want more rolls.



Price 35 UAH. Sales rise to 50 rolls.

Flour becomes more expensive

At each price, stalls offer fewer rolls.



Price 35 UAH. Sales fall to 30 rolls.

Price (UAH) is vertical. Rolls are horizontal. Dashed curves and filled dots show the new market.

What goes into a consumer price index?

PART OF A HOUSEHOLD'S SHOPPING



Food



Travel



Housing

A student who takes the bus every day may notice fares more than a family that cycles.

HOW THE INDEX WORKS

Statisticians track prices for many goods and services. Items that take up more household spending carry more weight.

The consumer price index (CPI) tracks the overall price level.

European Central Bank, What is inflation? Official spending weights are updated over time.

A TWO-ITEM BASKET

What changed on these receipts?

Marta prices the same two bread rolls and one bus ride each December.

DECEMBER, YEAR 0

2 rolls at 30 UAH

1 bus ride at 40 UAH

Total **100 UAH**

DECEMBER, YEAR 1

2 rolls at 40 UAH

1 bus ride at 40 UAH

Total **?**

IN PAIRS
2 MIN

Find the new total. How much has it risen in UAH and in percent?

The school basket shows the calculation with two items.

A 20 UAH increase on a 100 UAH basket

The bus fare stays at 40 UAH. The two rolls add 20 UAH to the bill.

ANNUAL CHANGE IN THE COST OF THIS BASKET

$$(120 - 100) \div 100 \times 100 = 20\%$$

For a country, the annual inflation rate is the percentage change in the overall price level from a year earlier.

European Central Bank, how a basket is used to calculate inflation.

THE SAME BASKET ONE YEAR LATER

Did prices rise faster in Year 2?

In Year 2, two rolls at 45 UAH and a bus ride at 42 UAH bring the total to 132 UAH.

COST EACH DECEMBER

YEAR 0		YEAR 1		YEAR 2
100 UAH	+20 UAH	120 UAH	+12 UAH	132 UAH

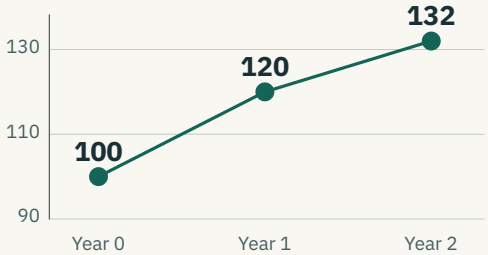
IN PAIRS
2 MIN

The first rise was 20%. Divide the second rise by its starting cost. Which year had the faster price growth?

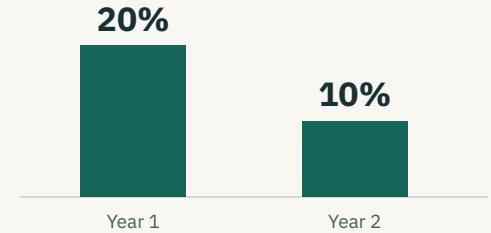
Classroom prices. All three observations use the same two rolls and one bus ride.

Prices rose more slowly in Year 2

COST OF THE SAME BASKET IN UAH



ANNUAL CHANGE IN THE BASKET COST



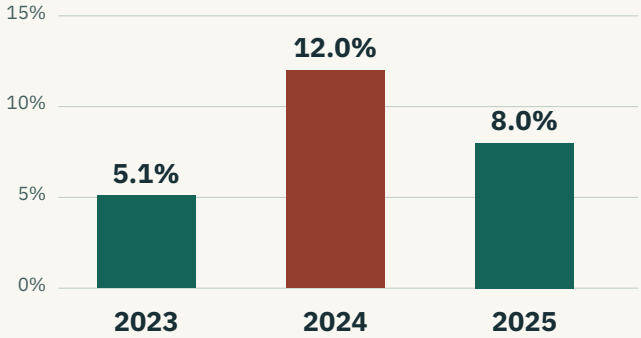
The rate fell by 10 percentage points. This slowdown is **disinflation**. When the rate rises, inflation **accelerates**.

A fall in the overall price level is deflation. St. Louis Fed, inflation, disinflation and deflation.

UKRAINIAN DATA

Inflation in Ukraine, 2023–2025

CONSUMER PRICES EACH DECEMBER COMPARED WITH THE PREVIOUS DECEMBER



From 2023 to 2024

Inflation rose by 6.9 percentage points. Poor harvests put pressure on food prices.

From 2024 to 2025

Inflation slowed. Larger harvests helped ease pressure on food prices.

National Bank of Ukraine, Inflation Reports for January 2025 and January 2026. National headline CPI.

AN AI DRAFT FOR THE SCHOOL PAPER

Would you publish this headline?

SCRIPTED AI REPLY

> **“Inflation fell from 12% to 8%, so prices in Ukraine fell by 4%.”**

The statistics office reports that prices in December 2025 were **8.0% higher than a year earlier** and **0.2% higher than in November**.

IN PAIRS
3 MIN

Write a new headline. Give the rise in prices and the change in the annual rate.

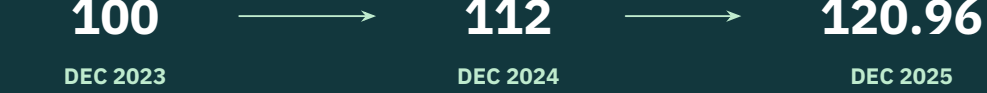
Classroom example. State Statistics Service of Ukraine, official releases for December 2024 and December 2025.

A HEADLINE THAT FITS THE DATA

Prices rose 8% as inflation slowed

In December 2025, consumer prices were 8% higher than a year earlier. The annual rate fell from 12% to 8%, a drop of four percentage points.

PRICE INDEX WITH DECEMBER 2023 SET TO 100



When checking an AI summary, open the original release. Match the dates and redo the calculation before sharing it.

Index calculated from the State Statistics Service releases for December 2024 and December 2025.

FINAL EXERCISE

Would a larger budget buy as much?

Marta has more money for the same two rolls and a bus ride.

AT THE SCHOOL MARKET	YEAR 1	YEAR 2
Money in her budget	**240 UAH**	**252 UAH**
Cost of one basket	**120 UAH**	**132 UAH**

ON YOUR OWN
2 MIN

How much would two baskets cost in Year 2? How far short is 252 UAH?

Use the same basket in both years.

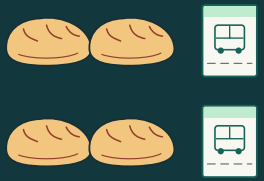
CLOSING DISCUSSION

Two baskets now cost 264 UAH

MARTA'S YEAR 2 BUDGET

252 UAH

Her budget rose by 5%. The basket price rose by 10%. She needs another 12 UAH to buy what she could buy before.



2 baskets at 132 UAH

End :-)

SOURCES

Reading and data

MONEY, PRICES AND INFLATION

Bank of England
What is money?

Federal Reserve Bank of St. Louis
How market prices form
Inflation, disinflation and deflation

European Central Bank
What is inflation?

OBSERVED UKRAINIAN CONSUMER PRICES

State Statistics Service of Ukraine
December 2024 price release
December 2025 price release

National Bank of Ukraine
January 2025 Inflation Report
January 2026 Inflation Report

The Ukrainian inflation series uses official data.